

GYAN SETU

Chapter: 4

Quadratic Equations

Introduction

A quadratic equation is an equation where the highest power of the variable is 2. It appears in many real-world problems involving areas, heights, motion, speeds, and optimization.

Example:

A hall has area 300 m². If breadth = x and length = $2x + 1$:

$$x(2x + 1) = 300$$

$$\rightarrow 2x^2 + x - 300 = 0$$

This is a quadratic equation.

Historically:

- Babylonians solved quadratic-type problems.
- Indians like Brahmagupta & Sridharacharya developed methods used today.
- Al-Khwarizmi (Persian mathematician) formalized quadratic solving.

Quadratic Equations

A quadratic equation is of the form:

$$ax^2 + bx + c = 0 \quad (a \neq 0)$$

Standard form is obtained by arranging terms in descending order of powers of x .

Examples:

$$- 2x^2 - 3x + 1 = 0$$

$$- x^2 - 55x + 750 = 0$$

$$- 1 - x^2 + 300 = 0 \text{ (rewrite as } -x^2 + 301 = 0)$$

Representing Situations:

Example:

Two persons have 45 marbles together. After losing 5 each, their product is 124.

Let John = $x \rightarrow$ Jivanti = $45 - x$

After losing 5:

$$(x - 5)(40 - x) = 124$$

$$\rightarrow x^2 - 45x + 324 = 0$$

Roots of a Quadratic Equation

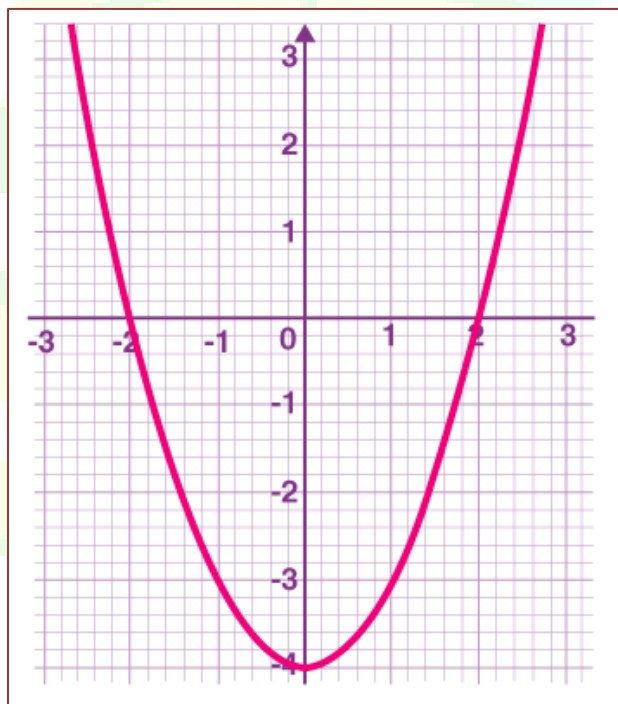
The values of x for which a quadratic equation is satisfied are called the roots of the quadratic equation.

If α is a root of the quadratic equation $ax^2+bx+c=0$, then $a\alpha^2+b\alpha+c=0$.

A quadratic equation can have two distinct real roots, two equal roots or real roots may not exist.

Graphically, the roots of a quadratic equation are the points where the graph of the quadratic polynomial cuts the x -axis.

Consider the graph of a quadratic equation $x^2-4=0$



Graph of a Quadratic Equation

In the above figure, -2 and 2 are the roots of the quadratic equation $x^2-4=0$.

Note:

- If the graph of the quadratic polynomial cuts the x-axis at two distinct points, then it has real and distinct roots.
- If the graph of the quadratic polynomial touches the x-axis, then it has real and equal roots.
- If the graph of the quadratic polynomial does not cut or touch the x-axis then it does not have any real roots.

Solving Quadratic Equations by Factorisation

Steps:

1. Multiply $a \times c$
2. Split b into two terms whose product = ac
3. Factorise into brackets
4. Set each factor = 0

Example:

$$2x^2 - 5x + 3 = 0$$

$$\text{Split } -5x \rightarrow -2x - 3x$$

Factorising:

$$(2x - 3)(x - 1) = 0$$

Roots:

$$x = 3/2, 1$$

Example:

$$6x^2 - x - 2 = 0$$

Factorised:

$$(3x - 2)(2x + 1) = 0$$

Roots:

$$x = 2/3, -1/2$$

Nature of Roots (Discriminant Method)

Quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Discriminant:

$$D = b^2 - 4ac$$

The value of D determines the nature of the roots.

Table: Nature of Roots Based on the Discriminant (D)

Value of Discriminant D	Nature of Roots	Explanation
$D > 0$	Two real and distinct roots	Square root of D is positive
$D = 0$	Two equal (coincident) real roots	Square root becomes zero, both roots are identical
$D < 0$	No real roots	Square root of a negative number is not real

Example:

$$2x^2 - 4x + 3 = 0$$

$$D = (-4)^2 - 4 \cdot 2 \cdot 3 = 16 - 24 = -8 \rightarrow \text{No real roots.}$$

Solving using Quadratic Formula when $D > 0$

Solve $2x^2 - 7x + 3 = 0$ using the quadratic formula.

(i) Identify the coefficients of the quadratic equation. $a = 2, b = -7, c = 3$

(ii) Calculate the discriminant, $b^2 - 4ac$

$$D = (-7)^2 - 4 \cdot 2 \cdot 3 = 49 - 24 = 25$$

$D > 0$, therefore, the roots are distinct.

(iii) Substitute the coefficients in the quadratic formula to find the roots

$$x = \frac{-(-7) \pm \sqrt{((-7)^2 - 4(2)(3))}}{2(2)}$$

$$x = \frac{7 \pm 5}{4}$$

$$x = 3 \text{ and } x = \frac{1}{2} \text{ are the roots.}$$

Solving Quadratic Equation when $D=0$

Let us take an example of quadratic equation $3x^2 - 2x + 1/3 = 0$.

Here, $a = 3$, $b = -2$ and $c = 1/3$

$$\begin{aligned}\text{Determinant, } D &= b^2 - 4ac = (-2)^2 - 4(3)(1/3) \\ &= 4 - 4 \\ &= 0\end{aligned}$$

Thus, the given equation has equal roots.

Hence the roots are $-b/2a$ and $-b/2a$, i.e., $1/3$ and $1/3$.

Solving Quadratic Equation when $D < 0$

Suppose the quadratic equation is $4x^2 + 3x + 5 = 0$

Comparing with the standard form of quadratic equation, $ax^2 + bx + c = 0$,

$a = 4$, $b = 3$, $c = 5$

By the formula of determinant, we know;

$$\begin{aligned}\text{Determinant (D)} &= b^2 - 4ac \\ &= (3)^2 - 4(4)(5) \\ &= 9 - 80\end{aligned}$$

$= -71 < 0$ So, $D < 0$ and hence the roots are complex (not real).

Summary

- Quadratic equation: $ax^2 + bx + c = 0$, $a \neq 0$.
- Roots satisfy $ax^2 + bx + c = 0$.
- Solve using factorisation or quadratic formula.
- Nature of roots depends on discriminant:
 - $D > 0 \rightarrow 2$ real distinct roots
 - $D = 0 \rightarrow 2$ equal roots
 - $D < 0 \rightarrow$ no real roots